MULTI-HAZARD EARLY WARNING AND DISASTER TECHNOLOGY

END-TO-END MHEWS RAIL → RISK-KNOWLEDGE STACK → OBSERVATION-TO-WARNING FIREWALL → DISSEMINATION REDUNDANCY → WARNING-TO-ACTION TEST → AGENCY OWNERSHIP MATRIX

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ● SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artifacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g1 and all prior artifacts unchanged

00

STAGE 00

END-TO-END MHEWS RAIL

END-TO-END MHEWS RAIL · 1 RISK KNOWLEDGE · 2 MONITOR · 3 AUTHORITATIVE WARNING · 4 DISSEMINATE · 5 PREPARE · 6 FEEDBACK

HAZARD / RISK / RESILIENCE CONCEPT

- 1 RISK KNOWLEDGE
- 2 MONITOR / OBSERVE / FORECAST

INSTITUTIONAL OR COMMUNITY RESPONSE

- 3 AUTHORITATIVE WARNING
- 4 DISSEMINATE → 5 PREPARE / ACT → 6 FEEDBACK

CAPACITY / STATUS / EVIDENCE LIMIT

- 1 RISK KNOWLEDGE
- 2 MONITOR / OBSERVE / FORECAST

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 4 DISSEMINATE → 5 PREPARE / ACT → 6 FEEDBACK.

01

STAGE 01

RISK-KNOWLEDGE STACK

RISK-KNOWLEDGE STACK · HAZARD + EXPOSURE + VULNERABILITY + CAPACITY · SPATIAL LAYERS + HISTORICAL · REAL-TIME DATA · WHO WHERE WHEN WHAT ACTION · COMMUNITY AND EVENT FEEDBACK

HAZARD + EXPOSURE + VULNERABILITY + CAPACITY

SPATIAL LAYERS + HISTORICAL / REAL-TIME DATA → DECISION → WHO WHERE WHEN WHAT ACTION → UPDATE → COMMUNITY AND EVENT FEEDBACK

HAZARD / RISK / RESILIENCE CONCEPT

- HAZARD + EXPOSURE + VULNERABILITY + CAPACITY
- SPATIAL LAYERS + HISTORICAL / REAL-TIME DATA

INSTITUTIONAL OR COMMUNITY RESPONSE

- DECISION → WHO WHERE WHEN WHAT ACTION
- UPDATE → COMMUNITY AND EVENT FEEDBACK

CAPACITY / STATUS / EVIDENCE LIMIT

- HAZARD + EXPOSURE + VULNERABILITY + CAPACITY
- SPATIAL LAYERS + HISTORICAL / REAL-TIME DATA

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DECISION → WHO WHERE WHEN WHAT ACTION; UPDATE → COMMUNITY AND EVENT FEEDBACK.

02

STAGE 02

OBSERVATION-TO-WARNING FIREWALL

OBSERVATION-TO-WARNING FIREWALL · QUALITY CONTROL · MODEL + EXPERT ANALYSIS · COMPETENT AUTHORITY · PUBLIC ACTION · NOT RAW DATA

SENSOR / REPORT → QUALITY CONTROL → MODEL + EXPERT ANALYSIS → FORECAST / ASSESSMENT → COMPETENT AUTHORITY → WARNING → PUBLIC ACTION → NOT RAW DATA

HAZARD / ACTOR / INSTRUMENT

- SENSOR / REPORT → QUALITY CONTROL
- MODEL + EXPERT ANALYSIS → FORECAST / ASSESSMENT

DECISIVE RISK OR MANDATE DISTINCTION

- COMPETENT AUTHORITY → WARNING
- PUBLIC ACTION → NOT RAW DATA

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- SENSOR / REPORT → QUALITY CONTROL
- MODEL + EXPERT ANALYSIS → FORECAST / ASSESSMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MODEL + EXPERT ANALYSIS → FORECAST / ASSESSMENT.

03

STAGE 03

DISSEMINATION REDUNDANCY

DISSEMINATION REDUNDANCY · CAP MESSAGE · LOCAL RELAY · PUBLIC ADDRESS · POWER + NETWORK + MANUAL PATH

CAP MESSAGE → SMS / CELL / APP / WEB

CAP MESSAGE → RADIO / TV / SIREN → LOCAL RELAY → VOLUNTEER / PUBLIC ADDRESS → BACKUP → POWER + NETWORK + MANUAL PATH

HAZARD / RISK / RESILIENCE CONCEPT

- CAP MESSAGE → SMS / CELL / APP / WEB
- CAP MESSAGE → RADIO / TV / SIREN

INSTITUTIONAL OR COMMUNITY RESPONSE

- LOCAL RELAY → VOLUNTEER / PUBLIC ADDRESS
- BACKUP → POWER + NETWORK + MANUAL PATH

CAPACITY / STATUS / EVIDENCE LIMIT

- CAP MESSAGE → SMS / CELL / APP / WEB
- CAP MESSAGE → RADIO / TV / SIREN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BACKUP → POWER + NETWORK + MANUAL PATH.

04

STAGE 04

WARNING-TO-ACTION TEST

WARNING-TO-ACTION TEST · ACTION FEASIBLE? · SHELTER AVAILABLE? · EARLY ACTION OR LAST-MILE FAILURE

RECEIVED? → UNDERSTOOD?

TRUSTED? → ACTION FEASIBLE? → ROUTE / TRANSPORT / SHELTER AVAILABLE? → RESULT → EARLY ACTION OR LAST-MILE FAILURE

HAZARD / RISK / RESILIENCE CONCEPT

- RECEIVED? → UNDERSTOOD?
- TRUSTED? → ACTION FEASIBLE?

INSTITUTIONAL OR COMMUNITY RESPONSE

- ROUTE / TRANSPORT / SHELTER AVAILABLE?
- RESULT → EARLY ACTION OR LAST-MILE FAILURE

CAPACITY / STATUS / EVIDENCE LIMIT

- RECEIVED? → UNDERSTOOD?
- TRUSTED? → ACTION FEASIBLE?

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ROUTE / TRANSPORT / SHELTER AVAILABLE?; RESULT → EARLY ACTION OR LAST-MILE FAILURE.

05

STAGE 05

AGENCY OWNERSHIP MATRIX

AGENCY OWNERSHIP MATRIX · RIVER FLOOD FORECAST · EARTH OBSERVATION · NDMA + STATE · INTEGRATION AND ACTION

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
IMD → WEATHER / CYCLONE	INCOIS → TSUNAMI	NDMA + STATE / LOCAL → INTEGRATION AND ACTION
CWC → RIVER FLOOD FORECAST	ISRO / NRSC → EARTH OBSERVATION / GIS	IMD → WEATHER / CYCLONE

HAZARD / ACTOR / INSTRUMENT

- IMD → WEATHER / CYCLONE
- CWC → RIVER FLOOD FORECAST

DECISIVE RISK OR MANDATE DISTINCTION

- INCOIS → TSUNAMI
- ISRO / NRSC → EARTH OBSERVATION / GIS

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- NDMA + STATE / LOCAL → INTEGRATION AND ACTION
- IMD → WEATHER / CYCLONE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NDMA + STATE / LOCAL → INTEGRATION AND ACTION.

06

STAGE 06

CAP AND SACHET BOUNDARY

CAP AND SACHET BOUNDARY · STRUCTURED MESSAGE FORMAT · NATIONAL ALERT PORTAL · DISSEMINATION LAYER · CHANNEL-FORMAT INTEROPERABILITY · DOES NOT SOLVE · HAZARD OWNERSHIP OR LOCAL READINESS

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
CAP → structured message format	SOLVES → channel-format interoperability	CAP → structured message format
SACHET → national alert portal / dissemination layer	DOES NOT SOLVE → hazard ownership or local readiness	SACHET → national alert portal / dissemination layer

HAZARD / ACTOR / INSTRUMENT

- CAP → structured message format
- SACHET → national alert portal / dissemination layer

DECISIVE RISK OR MANDATE DISTINCTION

- SOLVES → channel-format interoperability
- DOES NOT SOLVE → hazard ownership or local readiness

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- CAP → structured message format
- SACHET → national alert portal / dissemination layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DOES NOT SOLVE → hazard ownership or local readiness.

07

STAGE 07

GEOSPATIAL DECISION CYCLE

GEOSPATIAL DECISION CYCLE · FIELD DATA · GIS LAYERS · RISK MAP · EVENT IMAGE · ACCESS ASSESSMENT · RECOVERY MAP · SAFER SITING

SATELLITE / AERIAL / FIELD DATA

GIS LAYERS → RISK MAP / ROUTE / SHELTER → EVENT IMAGE → DAMAGE / ACCESS ASSESSMENT → RECOVERY MAP → SAFER SITING

RISK CONDITION / INPUT

- SATELLITE / AERIAL / FIELD DATA
- GIS LAYERS → RISK MAP / ROUTE / SHELTER

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- EVENT IMAGE → DAMAGE / ACCESS ASSESSMENT
- RECOVERY MAP → SAFER SITING

LAST-MILE / INCLUSION / OUTCOME LIMIT

- SATELLITE / AERIAL / FIELD DATA
- GIS LAYERS → RISK MAP / ROUTE / SHELTER

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GIS LAYERS → RISK MAP / ROUTE / SHELTER.

08

STAGE 08

SENSOR-HAZARD MATCHING

SENSOR-HAZARD MATCHING · DOPPLER RADAR · WEATHER STRUCTURE · SEA LEVEL · OCEAN RESPONSE · SEISMIC NETWORK · EARTHQUAKE PARAMETERS

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
DOPPLER RADAR → WEATHER STRUCTURE	BUOY / BPR → OCEAN RESPONSE	DOPPLER RADAR → WEATHER STRUCTURE
GAUGE → RAIN / RIVER / SEA LEVEL	SEISMIC NETWORK → EARTHQUAKE PARAMETERS	GAUGE → RAIN / RIVER / SEA LEVEL

HAZARD / ACTOR / INSTRUMENT

- DOPPLER RADAR → WEATHER STRUCTURE
- GAUGE → RAIN / RIVER / SEA LEVEL

DECISIVE RISK OR MANDATE DISTINCTION

- BUOY / BPR → OCEAN RESPONSE
- SEISMIC NETWORK → EARTHQUAKE PARAMETERS

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- DOPPLER RADAR → WEATHER STRUCTURE
- GAUGE → RAIN / RIVER / SEA LEVEL

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GAUGE → RAIN / RIVER / SEA LEVEL.

09

STAGE 09

EMERGING-TECHNOLOGY SAFEGUARDS

EMERGING-TECHNOLOGY SAFEGUARDS · LOCAL IMAGERY · AIRSPACE + SAFETY · UNCERTAINTY + OVERSIGHT · CROWD REPORT · LOCAL SIGNAL · VERIFICATION + PROVENANCE · PRIVACY + MINIMISATION

DRONE → local imagery	airspace + safety
AI → pattern / prioritisation	uncertainty + oversight
CROWD REPORT → local signal	verification + provenance
DATA → utility	privacy + minimisation

HAZARD / ACTOR / INSTRUMENT

- DRONE → local imagery
- airspace + safety
- AI → pattern / prioritisation

DECISIVE RISK OR MANDATE DISTINCTION

- uncertainty + oversight
- CROWD REPORT → local signal
- verification + provenance

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- DATA → utility
- privacy + minimisation
- MUST REMEMBER: An end-to-end early-warning chain runs observation → detection →...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: An end-to-end early-warning chain runs observation → detection →.

10

STAGE 10

FAILURE AND REDUNDANCY MAP

FAILURE AND REDUNDANCY MAP · SENSOR FAIL · POWER FAIL · NETWORK FAIL · AUTHORITY DELAY · FORMAT MISMATCH · ACCESS BARRIER · NO TRANSPORT

SENSOR FAIL

POWER FAIL → NETWORK FAIL → AUTHORITY DELAY

HAZARD / RISK / RESILIENCE CONCEPT

- SENSOR FAIL
- POWER FAIL
- NETWORK FAIL

INSTITUTIONAL OR COMMUNITY RESPONSE

- AUTHORITY DELAY
- FORMAT MISMATCH
- LANGUAGE / ACCESS BARRIER

CAPACITY / STATUS / EVIDENCE LIMIT

- NO TRANSPORT
- CONTROL → REDUNDANT TECHNICAL AND HUMAN PATHS
- CLOSE DISTINCTION: Observation is not forecast; forecast is not watch; watch is not...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Observation is not forecast; forecast is not watch; watch is not.

11

STAGE 11

TECHNOLOGY ANSWER SPINE

TECHNOLOGY ANSWER SPINE · MAP HAZARD + LEAD TIME · NAME COMPETENT AGENCY · TRACE DATA · ADD ACCESSIBLE ACTION + FEEDBACK · QUALIFY UNCERTAINTY PRIVACY AND OUTCOME EVIDENCE · EVIDENCE LIMIT · ATTRIBUTE THE COMPETENT AGENCY AND

MAP HAZARD + LEAD TIME → NAME COMPETENT AGENCY → TRACE DATA → FORECAST → WARNING → DISSEMINATION → ADD ACCESSIBLE ACTION + FEEDBACK → QUALIFY UNCERTAINTY PRIVACY AND OUTCOME EVIDENCE → EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Attribute the competent agency and

HAZARD / RISK / RESILIENCE CONCEPT

- MAP HAZARD + LEAD TIME → NAME COMPETENT AGENCY
- TRACE DATA → FORECAST → WARNING → DISSEMINATION

INSTITUTIONAL OR COMMUNITY RESPONSE

- ADD ACCESSIBLE ACTION + FEEDBACK
- QUALIFY UNCERTAINTY PRIVACY AND OUTCOME EVIDENCE

CAPACITY / STATUS / EVIDENCE LIMIT

- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Attribute the competent agency and...
- MAP HAZARD + LEAD TIME → NAME COMPETENT AGENCY

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Attribute the competent agency and...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Attribute the competent agency and.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · DISTRIBUTED BUT CENTRALLY VALIDATED ARCHITECTURE · THE INDIAN · AUTHORITY-CENTRALISATION TRADE-OFF · THE REQUIREMENT THAT ONLY · GIS'S THREE-PHASE INDIAN TRACK RECORD · VULNERABILITY MAPPING IN · PHAILIN'S "EXACT PREDICTION" CLAIM REQUIRES CAREFUL FRAMING

OPTIONAL RISK / RESILIENCE DEPTH

- Distributed but centrally validated architecture : the Indian
- CAUTION: Authority-centralisation trade-off : the requirement that only
- GIS's three-phase Indian track record : vulnerability mapping in
- CAUTION: Phailin's "exact prediction" claim requires careful framing :

MANDATE, WARNING OR OUTCOME LIMIT

- CAUTION: Impact-based forecasting is a genuine, only partially realised,
- CAUTION: Data redundancy and system vulnerability : VisionIAS notes
- CAUTION: Last-mile failure is not evenly distributed: households without
- CAUTION: Speed versus accuracy in dissemination is a structural trade-off

COMPARATIVE HAZARD / GOVERNANCE NUANCE

- NOT: The "technology deployed, therefore warning succeeded" fallacy.
- CAUTION: Interoperability across agencies (IMD, INCOIS, GSI, CWC, ISRO)
- CAUTION: VisionIAS does not detail the financing of India's warning
- Monitoring warning-system performance (lead time achieved, false

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels